UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
EASTER 2026 SEMESTER **EXAMINATION**

PROGRAM: BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

YEAR: 1 SEMESTER: 2

COURSE CODE: CSC1202

COURSE NAME: WEB AND MOBILE APPLICATION DEVELOPMENT

EXAMINATION TYPE: 100% PROJECT-BASED EXAM

PROJECT DURATION: APRIL 2026

TIME ALLOWED: TWO WEEKS

Examination Instructions

1. The general guidelines for the Uganda Christian University examination, as well as its academic & financial policies, apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.
2. This exam consists of a project that must be completed within ***two*** weeks.
   a. Assessment of the project shall be based on **five milestones**, evaluated during the duration of the project. **Each milestone** shall be evaluated out of **20 marks**.
   b. At the end of the project, the following SHALL be submitted on Moodle.
      i. A well-written project (Font: Trebuchet MS, 12Pts, 1.5 spacing, justified, aligned), **IEEE Referencing style**.
      ii. **Two links to GitHub** (Frontend and Backend), and these repositories should have evidence of collaboration and consistent individual commits.
3. This examination shall be attempted in collaborative teams. Every student is responsible for proving their contribution towards every milestone, and marks may be awarded to every student individually. This includes GitHub commits and pull requests.

PART A: PROJECT DESCRIPTION

Introduction

Collaborative learning is an important aspect of university education. Students often form informal study groups to discuss course materials, revise for examinations, and support each other in understanding challenging topics.

At Uganda Christian University, many study groups are formed through personal networks or class communication channels such as WhatsApp groups. While these approaches are useful, they often limit participation to students who already know each other. New students or those outside these networks may find it difficult to locate study groups related to their courses.

A simple digital platform that allows students to discover and organize study groups could improve collaboration and academic support across the university.

In this project, students are required to design and develop a web-based application called **"Student Study Group Finder."**

The system should enable students to create study groups for particular courses, allow other students to discover and join these groups, and provide tools for organizing study sessions.

Business Overview

The **Student Study Group Finder** will serve as a campus platform where students can connect with peers studying similar courses.

Students should be able to create groups dedicated to specific subjects, invite or accept members, and schedule study meetings. Other students should be able to browse available groups and join those that match their academic interests.

The system will support the following primary users:

a) Students

Students will be able to create accounts, join study groups, and participate in group discussions.

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b) Group Leaders

Students who create a group automatically become group leaders and will manage group activities and members.

c) System Administrator

An administrator will oversee the platform and monitor activity within the system.

Functional Requirements

1. User Management

The system should support:

a) Student registration and login

b) Secure authentication and session management

c) User profiles containing:

→ Name

→ Program of study

→ Year of study

2. Study Group Creation and Management

Students should be able to create study groups by providing:

→ Group name

→ Course name or code

→ Short description of the study focus

→ Meeting location (physical or online)

The creator of the group automatically becomes the **group leader**.
Group leaders should be able to:

→ Edit group information

→ Manage group members

→ Remove inactive members if necessary

3. Study Group Repository and Search

The system should provide a repository of study groups that students can browse.
Users should be able to search or filter groups by:

→ Course name
→ Faculty
→ Group title

Each group should display details, including the group description and the number of members.

4. Study Session Scheduling

Group leaders should be able to create study sessions for their group members.
Each study session should include:

→ Date
→ Time
→ Location or meeting link
→ Brief description of the session

Members should be able to view upcoming study sessions from their group dashboard.

5. Group Communication

The platform should allow members of a group to interact through short posts or comments.
This feature will allow students to:

→ Share announcements
→ Ask questions
→ Coordinate study meetings

6. Dashboard and Activity Overview

Students should have access to a dashboard showing:

→ Study groups, they belong to
→ Upcoming study sessions
→ Recently created groups

The administrator dashboard should provide information such as:

→ Total registered users
→ Total number of study groups
→ Most active courses

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Technical Requirements

1. Frontend Framework: **React, Vue.js, or Angular**.
2. Backend Framework: **Node.js (Express)**, **Django**, or **Laravel**.
3. Database: **MySQL**, **PostgreSQL**, or **MongoDB**.
4. Authentication: JWT-based secure login and role management.
5. Responsive design suitable for desktop and mobile.
6. Data validation, error handling, and a clean user interface.

Deliverables

Students must submit the following:

1. Source Code

Well-documented frontend and backend repositories hosted on **GitHub**.

2. Documentation

The project report should include:

- System architecture diagram
- ER diagrams
- User manual
- API documentation

3. Demonstration

A functional prototype should be demonstrated during project evaluation.

Key Notes

1. Students should focus on building a minimum viable product (MVP) that demonstrates the system's core features.
2. Collaboration between team members should be clearly visible through **Git commits and contribution logs**.
3. Code should be well-structured, modular, and easy to read.
4. During the final presentation, students should demonstrate both the **system's technical implementation and usability**.

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PART B: PROJECT-BASED ASSESSMENT GUIDELINES

S/N	Milestone Description	Marks
1	**MILESTONE ONE - Conceptualization** a) Define the problem, objectives, and scope. b) Provide system architecture, ER diagrams, and interface mockups. c) Explain technology choices.	20%
2	**MILESTONE TWO - Core Development** a) Implement user authentication, group creation, and database structure. b) Demonstrate basic group listing functionality.	20%
3	**MILESTONE THREE - Feature Development** a) Implement group membership, search functionality, and study session scheduling features.	20%
4	**MILESTONE FOUR - Implementation Quality** a) Demonstrate clean code practices, API validation, and responsive user interface design.	20%
5	**MILESTONE FIVE - Final Demonstration** a) Present the working system, explain architecture decisions, and respond to questions during the defense.	20%

TOTAL MARKS: 100%

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